

Robust Renewable Charging Station Planning with Spatio-temporal Flexibility and Continuous Endogenous Uncertainty

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Abstract—The planning of fast charging stations (CSs) not only satisfies the escalating charging demand of electric vehicles (EVs), but reshapes its spatio-temporal distribution. However, most existing CS planning studies neglect this endogenous feedback and treat charging demand as an exogenous input, resulting in inefficient infrastructure utilization, heightened grid strain and skewed energy evaluations. To address this gap, this article proposes a two-layer robust renewable CS (RCS) planning and operation model considering spatio-temporal flexibility and continuous decision-dependent uncertainty (DDU) of charging demand. In the first layer, an RCS-capacity-dependent charging demand transfer model is developed based on human mobility empirics, and the corresponding continuous DDU set is constructed. In the second layer, a carbon-aware robust RCS planning and operation model is formulated by integrating charging demand response, capacity-dependent demand uncertainty, and capacity-scaled renewable uncertainty. Since the continuous DDU sets introduce nonlinear and nonconvex couplings into the planning model, an approximation optimization model and a combined alternating direction method and convex-concave procedure algorithm are developed. Numerical studies based on real-world transportation and distribution networks demonstrate that the proposed model improves demand–service matching, reduces transportation-network cost, and achieves a better balance among infrastructure investment, grid operation, and low-carbon charging.

Index Terms—Electric vehicle, renewable charging station, decision-dependent uncertainty, robust multi-stage planning, linear programming relaxation, convex-concave procedure.

NOMENCLATURE

A. Abbreviation

EV	Electric vehicle
CS	Charging station
RCS	Renewable charging station
TN	Transportation network
PDN	Power distribution network
CPTN	Coupled power-transport network
SCP	Smart city planner
DR	Demand response
ESS	Energy storage system
PV	Photovoltaic
MILP	Mixed integer linear program
DIU	Decision-independent uncertainty
DDU	Decision-dependent uncertainty

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BUS	Budgeted uncertainty set
ADM	Alternating direction method
CCP	Convex-concave procedure
DC	Difference of convex
QC	Quadratic constraint

B. Sets

$\mathcal{N}_{pdn}, \mathcal{E}_{pdn}$	The sets of nodes and edges in the PDN.
$\mathcal{N}_{tn}$	The set of nodes in the TN.
A	The set of RCS candidates.
S	The set of typical operation scenarios.
$\mathcal{I}_t$	The set of operation intervals at stage t .
T	The set of planning stages.

C. Variables

$z_{i,t}^{cs}, \Delta z_{i,t}^{cs}$	Binary existence indicator and new investment decision of i -th RCS at stage t .
$y_{i,t}^{cs,pv,es}$	The total charger, PV and ESS capacities of i -th RCS in kW at stage t .
$\Delta y_{i,t}^{cs,pv,es}$	The expanded charger, PV and ESS capacities in kW of i -th RCS at stage t .
$P_{\tau,t,s}^{mg} / Q_{\tau,t,s}^{mg}$	Active and reactive power from main grid in kW at time τ , scenario s , stage t .
$e_{i,\tau,t,s}^{es}$	Energy level of i -th ESS in kWh at time τ , scenario s , stage t .
$P_{i,\tau,t,s}^{es+}, P_{i,\tau,t,s}^{es-}$	Charging and discharging power of i -th ESS in kW at time τ , scenario s , stage t .
$P_{i,\tau,t,s}^{pv}$	Dispatched power of i -th PV in kW at time τ , scenario s , stage t .
$P_{i,j,\tau,t}^{ch}$	Scheduled charging power from i -th TN node to j -th RCS in kW at time τ , stage t .
$P_{i,\tau,t}^{sh}$	Shifted charging power of i -th TN node in kW at time τ , stage t .

I. INTRODUCTION

A. Background and Motivation

THE utilization of EVs helps to combat global warming and achieve carbon neutrality in the TN. Recently, China has put a lot of effort into the EV promotion through a wide array of policy incentive (i.e., purchase-tax exemptions). As a result, the national EV stock has surpassed 35 million, with an annual growth rate approaching 30%. Shenzhen, as a global pioneer in EVs, hosts over 1.2 million EVs, representing roughly 25 % of urban registered vehicles [1].

The rapid transportation electrification drives a substantial increase in EV charging demand. To accommodate this surge, the SCP should design a coherent urban CS network to meet the

charging needs [2]. Meanwhile, on-site PVs and ESSs should be integrated to deliver green electricity, unlock decarbonization potentials of EVs, and mitigate the charging stress imposed on the PDN [3].

This paper focuses on urban fast CS planning with onsite PVs and ESSs. In practice, the urban fast EV charging demand is intrinsically stochastic and is influenced by long-term factors (i.e., the EV growth rate) and short-term factors (i.e., daily charging behaviors) [4]. The long-term factors are mainly determined by macro policy and considered as exogenous uncertainties in this paper. On the other hand, the short-term charging behaviors can be shaped by decision variables, such as siting and sizing of CSs, due to the spatial accessibility and on-site capacities of nearby CSs [5]. Neglecting such endogenous demand transfer may lead to supply-demand mismatch of EV charging and inefficient allocation of costly urban charging assets.

Beyond this CS capacity-dependent demand redistribution, public fast charging also exhibits spatio-temporal flexibility, because a portion of charging demand can be shifted across time in response to service conditions and DR incentives [6]. Spatial flexibility alters the locations at which charging load is concentrated, while temporal flexibility changes the timing of that load. Both of them are crucial for improving charging service matching, alleviating grid stress, and enhancing renewable utilization [7]. Consequently, a well-designed urban CS layout with consideration of spatio-temporal charging flexibility can guarantee high-quality service and indirectly influence drivers' charging preferences, thereby fostering a more balanced, resilient, and sustainable urban CPTN.

This paper aims to provide high-quality charging service as well as achieving low-carbon operation in the urban CPTN via RCS planning and scheduling considering the influence of planning decisions on the EV charging demand transfer and spatio-temporal charging flexibility. Generally, this can be described as a multi-period planning model and is the main research work of this paper.

B. Literature Review

Planning of CSs imposes a coordination requirement for the CPTN and previous studies for CS planning approaches are reviewed in this subsection. Key features of these studies are summarized in Table I.

The authors in [8] propose a coordinated planning model of multiple facilities, including electric power lines, ESSs and fast CSs. However, considering the high cost and low flexibility of power line expansion, recent studies have highlighted the advantages of RCS with on-site PVs and ESSs. For instance, [9] studies the planning strategy of EV fast CSs equipped with ESSs in a competitive environment. Besides, a RCS planning method based on fuzzy inference system is proposed in [10]. A data-driven predict-then-optimize planning framework is designed in [11], where the charging demand is predicted first and the planning decisions are made based on the predicted values. Recently, a multi-objective optimization strategy for CS planning is proposed in [12] that jointly optimizes grid stability, traffic congestion, and infrastructure costs. However, these planning models ignore the uncertainties of charging demand and renewables.

To deal with these uncertainties, [13] presents a data-driven robust planning model for fast CSs fully powered by on-site

PVs. In [14], a probabilistic MILP model is proposed to integrate renewables and ESSs into fast CSs. A joint planning and operation model of CSs considering users' participation willingness is explored in [15]. The authors [16] present a stochastic planning model for RCSs to minimize the costs of network investment, energy losses, and waiting time of charging services. Further, a queueing mechanism is proposed to account for the charging demand distribution and an iterative heuristic algorithm is designed to arrange the charging ports across the CSs in [17]. A fast CS planning model considering long-term EV adoption and short-term quality of service (QoS) uncertainties is proposed in [18]. A multistage robust planning optimization for RCSs is explored in [19] and an improved robust dual dynamic program method is proposed to solve the planning model. The authors in [20] propose a two-stage scheduling model to coordinate both fixed and mobile CSs, which leverages extended graph model to capture service provision characteristics and value function approximation to address demand uncertainty. Although these researches make contributions to RCS planning in different settings, they characterize the charging demand as a DIU, whose realizations are only related to exogenous factors such as typical operation days. In practice, the siting and sizing of RCSs substantially affect the EV adoption [21] and charging behaviors of drivers [22].

The problems with DDU have been widely explored in many areas. In [21], a decision-dependent distributionally robust (D3R) fast CS planning model is developed where the decision-dependent ambiguity sets are constructed for the uncertain yearly EV adoption rate. Similarly, the data-driven D3R planning models are applied in hydrogen supply infrastructure in [22] and logistics facility location in [23] but lead to low computational efficiency. The impact of fast CS allocation on the willingness of drivers to deviate from their paths for refueling are studied in [24] and a DDU set is constructed to quantify the uncertain path flow. [25] considers the wind power uncertainty to be dependent on the investment decisions of wind plants and the numeric studies prove more cost-effective results of the DDU method. A robust microgrid dispatch model with endogenous uncertainty is proposed in [26] where the connection states of renewables influence the size and dimension of the uncertainty set.

Nevertheless, most existing studies model DDU using only binary decision variables like generator connection states and facility locations. Continuous DDU modeling delivers higher accuracy and flexibility for endogenous uncertainty-decision variable relations yet raises computational difficulty [27]. In urban CPTN EV charging demand research, prior work only examines fast charging station (CS) siting (binary variable), neglecting CS capacity impacts [28]. CS planning has diminishing marginal effects on EV growth: sufficient CS coverage makes EV growth user-driven. Meanwhile, physics-informed prediction proves high-capacity public CSs attract more EV drivers and handle larger charging loads [29]. Therefore, how to model the relationship between uncertain charging demand and CS sizing decisions, and efficiently solve the continuous DDU optimization problem is still underexplored.

TABLE I SUMMARY OF PREVIOUS STUDIES ON CS PLANNING

Ref.	Planning structure	Onsite units	Uncertainty modeling	Objective (Minimization)	Solution method
[8]	Coordinated planning	EV Charger, CPTN lines.	None	Costs of CPTN lines and CS planning, energy consumption and detour of users	Linearization method and Lagrangian relaxation
[9],[11]	Competitive planning	EV Charger, ESS.	None	Costs of energy consumption, siting and sizing of CSs, minus service revenue	Iterative searching
[10],[12]	Coordinated planning	EV Charger, ESS, PV.	None	Costs of siting and sizing of RCSs, energy consumption and carbon emissions	Multi-objective natural aggregation algorithm
[13],[14]	Coordinated planning	EV Charger, ESS, PV.	Stochastic DIU	Penalty of CS services accessibility and costs of CS siting and energy consumption	Monte Carlo + MILP
[15],[17]	Coordinated planning	EV Charger.	Stochastic DIU	Costs of siting of CSs, EV battery degradation and range anxiety	K-means clustering siting and heuristic dispatch
[16]	Coordinated planning	EV Charger, ESS, PV.	Stochastic DIU	Network investment cost, energy losses, and queue waiting time	Heuristic natural aggregation algorithm
[18]	Coordinated planning	EV Charger, ESS, PV.	Markov Process	Costs of siting and sizing of RCSs, energy consumption and detour of users	Multi-agent deep reinforcement learning
[19]	Coordinated planning	EV Charger, ESS, PV.	Robust DIU	Costs of siting and sizing of RCSs, energy consumption and detour of users	Robust dual dynamic programming
[20]	Coordinated planning	EV Charger, ESS.	Markov Process	Cost of mobile CS operation and degradation minus service revenue	Approximate dynamic programming
[21],[22]	Coordinated planning	EV Charger, ESS, PV.	Robust binary DDU	Costs of siting and sizing of RCSs, energy consumption and unsatisfied charging	McCormick envelope + MILP
This paper	Coordinated planning	EV Charger, ESS, PV.	Robust continuous DDU	Costs of siting and sizing of CSs, energy consumption, detour and shifted charging	Customized ADM-CCP algorithm + MILP

C. Contributions

To fill the gap of urban RCS planning with continuous DDU, firstly, we explicitly quantify the relationship between CS capacities and uncertain EV charging demand. Then, a robust RCS planning and scheduling model with continuous DDU is proposed. To solve the proposed model efficiently, we derive the linear program approximation model and design a customized algorithm to deal with nonconvexity introduced by continuous DDU. Finally, a real-world CPTN is used to validate the effectiveness of the proposed model and method in case studies. The key contributions are summarized as follows.

1) **Robust RCS planning model with continuous DDU.** We first establish a CS-capacity dependent charging demand transfer model based on human mobility empirics. Besides, the budgeted continuous DDU set is constructed to quantify the deviation of charging demand from the empirical value. Furthermore, based on the above DDU model, a robust carbon-aware RCS planning considering spatio-temporal charging flexibility and continuous DDU of charging demand is further developed. The blocked pricing mechanisms are designed for carbon emission and traveling cost to achieve carbon and QoS-aware planning of RCSs. Meanwhile, the scheduling flexibility of EVs is also considered through the incentive-based DR program.

2) **Improved algorithm combined with ADM and CCP.** To solve the proposed planning model with continuous DDU, the reformulation based on strong duality and linear program approximation is first derived. Moreover, to tackle the bilinear term introduced by continuous DDU, a customized algorithm combined with ADM and CCP is developed with the proof of convergence guarantee.

3) **Real-world CPTN validation and performance evaluation.** We conduct case studies based on a real-world CPTN to validate the proposed model and algorithm. The results demonstrate that the proposed continuous DDU

formulation improves demand-service spatial matching and reduces system cost, while the customized ADM-CCP algorithm achieves high-quality solutions with improved computational efficiency.

II. ROBUST PLANNING AND SCHEDULING MODEL OF RCS WITH DECISION DEPENDENT CHARGING DEMAND TRANSFER

This section proposes a two-layer charging demand transfer and charging service completion framework by optimal planning of RCSs, which is illustrated in Fig.1. In layer 1, the *decision dependent charging demand transfer* (DDCDT) model is established to quantify the impact of CS planning on charging demand redistribution. In layer 2, the *carbon-aware robust CS planning and scheduling model* is elaborated to optimally completing the charging demand with continuous DDU. Additionally, this paper adopts the perspective of a SCP, under which candidate CSs are cooperatively planned as coordinated infrastructure assets, rather than modeled as self-interested market competitors, which is consistent with most existing literature [13]-[22]. Accordingly, although expanding one CS may change the optimal capacities of other stations, this interaction is driven by system-level charging demand redistribution and infrastructure coordination, rather than by market competition.

A. Decision Dependent Charging Demand Transfer and Uncertainty in Layer 1

Firstly, we consider the base value of total daily charging demand in the TN at stage t as an exogenous parameter denoted by $\bar{d}_t^{base}$, since it is mainly influenced by the macro policy [28] under the demand-supply balance situation.

Secondly, the practical charging demand often reaches peak during low charging price period and grows larger with a higher CS capacity in Shenzhen [29], which makes the existing linear model between charging demand and traffic flow at user equilibrium in [19] invalid. Thereby, the spatiotemporal

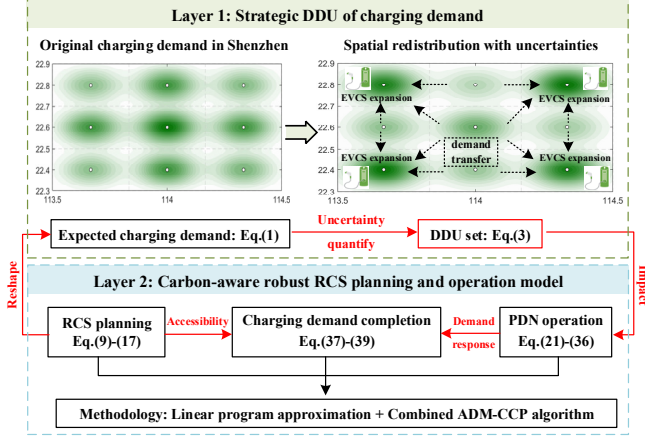


Fig. 1. The illustration of two-layer charging demand transfer and charging service completion framework.

distribution of charging demand is mainly influenced by charging price and CS capacity rather than traffic flow. Additionally, we will not explicitly optimize route flows or user-equilibrium traffic patterns in the TN constraints in Section II.C. Instead, the impact of transportation conditions is captured through effective inter-node distance, accessibility-based demand transfer, and the detour-related service cost in (8). This abstraction is adopted because the focus of this work is long-term station planning and charging-demand redistribution, rather than real-time route choice equilibrium.

To avoid ambiguity, we distinguish two timescales in this paper. The index $t \in T$ denotes the long-term planning stage (or planning period), at which infrastructure expansion decisions are made. For each planning stage t , the index $\tau \in \mathcal{T}_t$ denotes the short-term operation time interval used to model charging demand realization and system operation under typical days. The time span of each stage in the proposed model can be adjusted according to the practical needs of the smart city planner. In this paper, the timescale of “stage” is one year, and that of “operation time” is one hour.

Based on the above preliminaries, the expansion of CS at node i will transfer the charging demand from the original nodes to node i , which is defined as DDCDT. To quantify the influence of RCS planning decisions on the spatial redistribution of charging demand, we first define $\mathbf{M}^{dist}$ as the *distance matrix* of TN, with its element d_{ij} being the effective distance between node i and j . Besides, let $\mathbf{W}^{dist} = [\omega_{ij}^{dist}] \in R^{|\mathcal{V}_m| \times |\mathcal{V}_m|}$ as the symmetric *mobility-consistent affinity matrix* with its diagonal element $\omega_{ii}^{dist} = 0$, $\mathbf{D}^{dist} = \text{diag}(\mathbf{W}^{dist} \mathbf{e})$ as a generalized *degree matrix*, $\mathbf{L}^r = \mathbf{D}^{dist} - \mathbf{W}^{dist}$ as a *demand transfer Laplacian matrix*, and $\mathbf{e}$ as the column vector of all ones. Then, the transferred charging demand is modeled by a linear function of RCS capacity and is described in the compact form as follows:

$$\bar{\mathbf{d}}_{\tau,t}^{ch}(\mathbf{y}_t^{cs}) = \bar{\mathbf{d}}_{\tau,t}^{ch,0} + \alpha_{\tau}^r \mathbf{L}^r \mathbf{y}_t^{cs}; \forall \tau \in \mathcal{T}_t, t \in T, \quad (1)$$

where $\bar{\mathbf{d}}_{\tau,t}^{ch}, \bar{\mathbf{d}}_{\tau,t}^{ch,0} \in R^{|\mathcal{V}_m|}$ are the transferred and original nodal charging demand vectors at time τ and stage t . $\mathbf{y}_t^{cs} \in R^{|\mathcal{V}_m|}$ is the

planned capacity vector of RCSs at stage t . $\alpha_{\tau}^r > 0$ is the scaling factor. Since $\mathbf{e}^T \mathbf{L}^r = 0$, we have $\mathbf{e}^T \bar{\mathbf{d}}_{\tau,t}^{ch} = \mathbf{e}^T \bar{\mathbf{d}}_{\tau,t}^{ch,0}$, which preserves the total demand at each period. Meanwhile, to generate $\mathbf{W}^{dist}$, we parameterize node-pair affinity coefficients by the tempered power law of effective distance based on human mobility empirics in [30]. Specifically, $\mathbf{W}^{dist} = [\omega_{ij}^{dist}]$ is calculated by:

$$\omega_{ij}^{dist} = \left(1 + d_{ij} / \tau^{dist}\right)^{-\beta^{dist}} \exp(-d_{ij} / \kappa), \forall i \neq j \quad (2)$$

where $\beta^{dist}, \tau^{dist} > 0$ control the near-field decay and $\kappa > 0$ tempers the far tail. These parameters can be determined based on the practical system [30]. Eq.(2) preserves non-negligible medium/long range interactions that a pure exponential might over-suppress [30]. It is worth noting that d_{ij} in Eq.(2) should be interpreted as a generalized impedance metric for macroscopic spatial interactions. In megacities with severe traffic congestion, it can be replaced by congestion-adjusted generalized travel impedance when such data are available. In this study, a distance-based d_{ij} is used as a planning-stage proxy rather than a real-time traffic state variable. It captures the long-term tendency of demand transfer induced by station accessibility and capacities, while maintaining model tractability and consistency with the mobility-based transfer formulation. Based on Eq.(1) and (2), the RCS capacity-dependent charging demand is modeled.

An illustrative example is given as follows. Consider a 3-node TN with original charging demand $\bar{\mathbf{d}}_t^{ch,0} = [100, 100, 100]$ kWh at each node and $\bar{d}_t^{base} = 300$ kWh. The distance between each node is 5 kilometers. The parameters of Eq.(2) is set by: $\beta^{dist} = 1.8, \tau^{dist} = 0.8, \kappa = 350$ based on results in [30]. Assume the planning decision $\mathbf{y}_t^{cs} = [200, 100, 0]$ kWh to meet the total demand. Based on DDCDT model, the transferred charging demand becomes $\bar{\mathbf{d}}_t^{ch} = [108, 100, 92]$, indicating 8% demand transfer between node 1,3 and total demand unchanged during the practical operation period.

Finally, to make the planning and scheduling decisions more reliable, the charging demand uncertainty in the operation period needs to be quantified. In this article, the following DDU set is constructed for transferred charging demands:

$$\Xi_t^{ch}(\mathbf{y}_t^{cs}) := \left\{ \begin{array}{l} \xi_{\tau,t}^{ch} \in R^{|\mathcal{V}_m| \times |\mathcal{T}_t|}: \\ -\alpha_{i,\tau}^{ch} \bar{d}_{i,\tau,t}^{ch}(\mathbf{y}_t^{cs}) \leq \xi_{i,\tau,t}^{ch} \leq \alpha_{i,\tau}^{ch} \bar{d}_{i,\tau,t}^{ch}(\mathbf{y}_t^{cs}) \\ \sum_{i \in \mathcal{V}_m} |\xi_{i,\tau,t}^{ch}| \leq \Gamma_{\tau,t}^{ch} \\ \Gamma_{\tau,t}^{ch} = \beta_{\tau}^{ch} \sum_{i \in \mathcal{V}_m} \bar{d}_{i,\tau,t}^{ch}(\mathbf{y}_t^{cs}) \end{array} \right\} \quad (3)$$

where $\alpha_{i,\tau}^{ch}$ is the deviation factor at node i time τ . The uncertainty budget $\Gamma_{\tau,t}^{ch}$ models the spatial correlation of $\xi_{i,\tau,t}^{ch}$ to avoid over-conservativeness. Then, the DDCDT model with

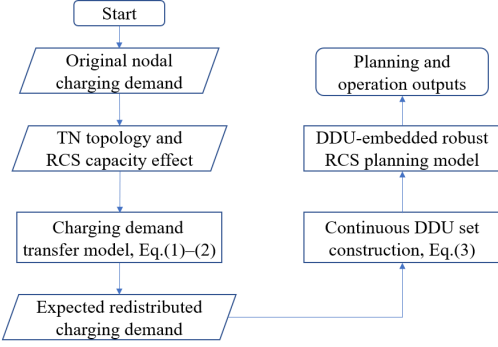


Fig. 2. Flowchart of the proposed DDCDT model with DDU set.

DDU set is formulated by Eq.(1)-(3), and will be embedded in the robust RCS planning model in Layer 2. Finally, an illustrative flow chart is provided in Fig. 2.

B. Objective Function of Layer 2

This paper adopts the perspective of a SCP, where charging infrastructure planning and system operation are jointly optimized to minimize the overall social cost throughout all planning periods, which can be described as follows:

$$V = \min \sum_{t \in T} (1 + \gamma_{IR})^{1-t} (C_t^{inv} + C_t^{ope,pn} + C_t^{ope,ce} + C_t^{ope,tn}) \quad (4)$$

$$C_t^{inv} = \sum_{a \in A} (c_{a,t}^{build} \Delta z_{a,t}^{cs} + c_{a,t}^{charger} \Delta y_{a,t}^{cs} + c_{a,t}^{pv} \Delta y_{a,t}^{pv} + c_{a,t}^{es} \Delta y_{a,t}^{es}) \quad (5)$$

$$C_t^{ope,pn} = \sum_{s \in S} D_s \sum_{\tau \in \mathcal{T}_t} \left(c_{\tau,t,s}^{mg} P_{\tau,t,s}^{mg} + c_{\tau,t}^{es} \sum_{a \in A} P_{a,\tau,t,s}^{es-} + c_{\tau,t}^r \sum_{a \in A} (\bar{\lambda}_{a,\tau,t,s}^{pv} y_{a,t}^{pv} - P_{a,\tau,t,s}^{pv}) \right) \quad (6)$$

$$C_t^{ope,ce} = \sum_{s \in S} D_s \max_{k=1,\dots,K} \left(c_{k,t}^{ce} \sum_{\tau \in \mathcal{T}_t} CI_{\tau,t,s}^{mg} P_{\tau,t,s}^{mg} + b_{k,t}^{ce} \right) \quad (7)$$

$$C_t^{ope,tn} = \sum_{s \in S} D_s \sum_{\tau \in \mathcal{T}_t} \sum_{i \in \mathcal{N}_m} \left(\max_{k=1,\dots,K} \left(\lambda_k^{dist} \sum_{a \in A} d_{ia} P_{i,a,\tau,t}^{ch} + b_k^{dist} \right) + \lambda^{inc} (P_{i,\tau,t}^{sh})^+ \right) \quad (8)$$

Eq.(5) is the investment cost of RCSs including construction expenses for new RCSs and the capacity expansion costs for chargers, PVs and ESSs, with prices $c_{a,t}^{build}$, $c_{a,t}^{charger}$, $c_{a,t}^{pv}$, $c_{a,t}^{es}$, respectively. γ_{IR} is the annual interest rate. $\Delta z_{a,t}^{cs}$, $\Delta y_{a,t}^{cs}$, $\Delta y_{a,t}^{pv}$, $\Delta y_{a,t}^{es}$ are planning variables modeled in Eq.(9)-(15). Eq.(6) is the operation cost of PDN including energy purchase cost from main grid, ESS degradation cost and PV curtailment cost with prices $c_{\tau,t,s}^{mg}$, $c_{\tau,t,s}^{es}$, $c_{\tau,t}^r$, respectively. D_s is the number of typical day s . Carbon emission cost of energy consumption from the main grid is calculated by Eq.(7) and the TN operation cost with detour cost and charging shift incentive by Eq.(8). $CI_{\tau,t,s}^{mg}$ is the carbon intensity (CI) of main grid. $(x)^+$ means $\max(x, 0)$. The blocked pricing mechanism is designed for both carbon emission (with blocked price $c_{k,t}^{ce}$ and bias $b_{k,t}^{ce}$) and traveling for charging (with blocked price λ_k^{dist} and bias b_k^{dist}), respectively. The detour cost modeled in (8) is introduced to capture the service disutility caused by assigning

charging demand to stations with higher travel burden. Empirical studies on public fast charging choices have shown that EV users are sensitive to detour distance and generally prefer stations requiring shorter detours [31]. Additionally, it should be noted that (8) is not intended to represent an exact individual-level route-choice utility. Instead, because the planning model is formulated at the aggregated charging demand level, the term $(d_{ia} P_{i,a,\tau,t}^{ch})$ is used to represent the demand-weighted detour burden. Moreover, while many existing studies (e.g., [32]) use a single linear detour term, the maximum of multiple linear functions in (8) is adopted to provide a more flexible penalty for larger detour burdens. This idea is similar to that in [33], which gives a piecewise linear approximation of an increasing convex service disutility. The illustration of the proposed detour cost function is shown in Fig. 3. With the proposed mechanisms, the higher service quality of RCS planning can be achieved.

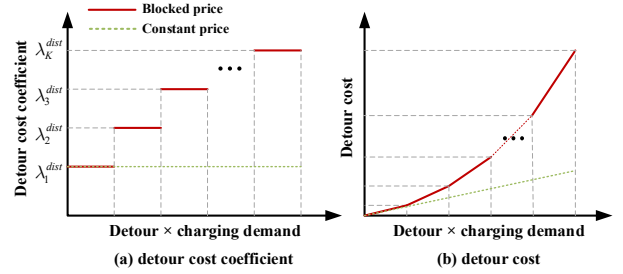


Fig. 3. The illustration of piecewise linear detour cost model with the maximum of multiple linear functions.

Besides, $P_{i,\tau,t}^{sh}$ is the temporal shifted power from the original charging demand considering EV charging flexibility and defined in Eq.(38). λ^{inc} is the incentive price for the DR.

C. System Constraints of Layer 2

The proposed coordinated planning and operation model is subject to the following constraints. 1) Multi-period planning constraints which facilitate logical planning decisions:

$$z_{a,t}^{cs} \in \{0,1\}, \forall a \in A; z_{a,t}^{cs} = 0, \forall a \notin A \quad (9)$$

$$z_{a,t}^{cs} \geq z_{a,t-1}^{cs}; \Delta z_{a,t}^{cs} = z_{a,t}^{cs} - z_{a,t-1}^{cs}, \forall a \in A, \forall t \in T \quad (10)$$

$$y_{a,t}^{cs} \geq y_{a,t-1}^{cs}; \underline{y}_{a,t}^{cs} z_{a,t}^{cs} \leq y_{a,t}^{cs} \leq \bar{y}_{a,t}^{cs} z_{a,t}^{cs} \quad (11)$$

$$\Delta y_{a,t}^{cs} = y_{a,t}^{cs} - y_{a,t-1}^{cs}; \Delta \underline{y}_{a,t}^{cs} \leq \Delta y_{a,t}^{cs} \leq \Delta \bar{y}_{a,t}^{cs} \quad (12)$$

$$y_{a,t}^{pv} \geq y_{a,t-1}^{pv}; \underline{y}_{a,t}^{pv} z_{a,t}^{cs} \leq y_{a,t}^{pv} \leq \bar{y}_{a,t}^{pv} z_{a,t}^{cs} \quad (13)$$

$$\Delta y_{a,t}^{pv} = y_{a,t}^{pv} - y_{a,t-1}^{pv}; \Delta \underline{y}_{a,t}^{pv} \leq \Delta y_{a,t}^{pv} \leq \Delta \bar{y}_{a,t}^{pv} \quad (14)$$

$$y_{a,t}^{es} \geq y_{a,t-1}^{es}; \underline{y}_{a,t}^{es} z_{a,t}^{cs} \leq y_{a,t}^{es} \leq \bar{y}_{a,t}^{es} z_{a,t}^{cs} \quad (15)$$

$$\Delta y_{a,t}^{es} = y_{a,t}^{es} - y_{a,t-1}^{es}; \Delta \underline{y}_{a,t}^{es} \leq \Delta y_{a,t}^{es} \leq \Delta \bar{y}_{a,t}^{es} \quad (16)$$

$$z_{a,0}^{cs} = 0, y_{a,0}^{cs} = 0, y_{a,0}^{pv} = 0, y_{a,0}^{es} = 0 \quad (17)$$

Eq.(9) and (10) model the binary existence indicator of RCS $z_{a,t}^{cs}$. Eq. (11)-(16) dictate that capacity expansions of chargers, PVs and ESSs are only made for constructed RCSs and should be limited within certain ranges. Eq.(17) specifies that the initial states of $z_{a,0}^{cs}$, $y_{a,0}^{cs}$, $y_{a,0}^{pv}$, $y_{a,0}^{es}$ should be zero. It should be noted that, in this paper, $y_{a,t}^{cs}$ denotes the aggregated station-level

charging capacity for strategic planning, rather than the exact number of chargers. This representation is adopted because the service requirement in the planning level is characterized by aggregated charging demand energy, rather than by the number of charging sessions and their charger-level power assignments. The system-level planning is directly coupled with aggregated station power, while charger count alone cannot uniquely characterize service capability without fixing charger ratings and vehicle-side charging acceptance. This representation is also consistent with existing literature [18]-[21].

Besides, the charging demand and loads increase with expected growth rates (denoted by $\phi_{i,\tau,t}^{ch}, \phi_{i,\tau,t}^L$) at each stage:

$$\bar{d}_{i,\tau,t}^{ch,0} = (1 + \phi_{i,\tau,t}^{ch}) \bar{d}_{i,\tau,t}^{ch,0}, \quad (18)$$

$$\bar{P}_{i,\tau,t}^L = (1 + \phi_{i,\tau,t}^L) \bar{P}_{i,\tau,t}^L, \quad (19)$$

$$\bar{Q}_{i,\tau,t}^L = (1 + \phi_{i,\tau,t}^L) \bar{Q}_{i,\tau,t}^L. \quad (20)$$

2) PDN operation constraints: The distribution flow model is adopted to describe the power flow constraints of the PDN.

$$p_{j,\tau,t} = \sum_{k:j \rightarrow k} P_{jk,\tau,t} - \sum_{i:i \rightarrow j} (P_{ij,\tau,t} - r_{ij} l_{ij,\tau,t}), \forall j \in \mathcal{N}_{pdn}, \tau \in \mathcal{T}_t \quad (21)$$

$$q_{j,\tau,t} = \sum_{k:j \rightarrow k} Q_{jk,\tau,t} - \sum_{i:i \rightarrow j} (Q_{ij,\tau,t} - x_{ij} l_{ij,\tau,t}) \quad (22)$$

$$v_{j,\tau,t} = \left(v_{i,\tau,t} - 2(r_{ij} P_{ij,\tau,t} + x_{ij} Q_{ij,\tau,t}) + (r_{ij}^2 + x_{ij}^2) l_{ij,\tau,t} \right), \forall (i,j) \in \mathcal{E}_{pdn} \quad (23)$$

$$\| [2P_{ij,\tau,t} \quad 2Q_{ij,\tau,t} \quad l_{ij,\tau,t} - v_{i,\tau,t}] \|_2 \leq l_{ij,\tau,t} + v_{i,\tau,t} \quad (24)$$

where $P_{ij,\tau,t} / Q_{ij,\tau,t}$ is the power flow from node i to j .

$v_{j,\tau,t} = |V_{j,\tau,t}|^2$ and $l_{ij,\tau,t} = |I_{ij,\tau,t}|^2$ are defined as the squared voltage and current magnitudes. r_{ij}, x_{ij} are the impedance on the line $(i,j) \in \mathcal{E}_{pdn}$. Meanwhile, the equality constraints of injection power at node j are given by

$$p_{j,\tau,t,s} = P_{j,\tau,t,s}^{mg} + P_{j,\tau,t,s}^{pv} - \bar{P}_{j,\tau,t,s}^L - \sum_{i \in \mathcal{N}_m} P_{i,j,\tau,t,s}^{ch} - P_{j,\tau,t,s}^{es} \quad (25)$$

$$q_{j,\tau,t,s} = Q_{j,\tau,t,s}^{mg} - \bar{Q}_{j,\tau,t,s}^L, \forall j \in \mathcal{N}_{pdn}, \tau \in \mathcal{T}_t, s \in S. \quad (26)$$

where $\bar{P}_{j,\tau,t,s}^L, \bar{Q}_{j,\tau,t,s}^L$ are the active and reactive load power. The nodal voltage and branch current should be constrained within the acceptable ranges, which are modeled by:

$$V_{\min}^2 \leq v_{j,\tau,t} \leq V_{\max}^2 \quad (27)$$

$$l_{ij,\tau,t} \leq I_{\max}^2. \quad (28)$$

Besides, the power purchase from the main grid should be limited within a feasible range as follows:

$$\underline{P}_{\tau,t}^{mg} \leq P_{\tau,t,s}^{mg} \leq \bar{P}_{\tau,t}^{mg}; \forall \tau \in \mathcal{T}_t, s \in S \quad (29)$$

$$\underline{Q}_{\tau,t,s}^{mg} \leq Q_{\tau,t,s}^{mg} \leq \bar{Q}_{\tau,t,s}^{mg}. \quad (30)$$

The operational constraints of installed PVs and ESSs are:

$$0 \leq P_{a,\tau,t,s}^{pv} \leq \bar{\lambda}_{a,\tau,t,s}^{pv} y_{a,t}^{pv}; \forall a \in A, \tau \in \mathcal{T}_t, s \in S \quad (31)$$

$$0 \leq e_{a,\tau,t,s}^{es} \leq y_{a,t}^{es} \quad (32)$$

$$0 \leq P_{a,\tau,t,s}^{es+} \leq \gamma^{ch} y_{a,t}^{es}; 0 \leq P_{a,\tau,t,s}^{es-} \leq \gamma^{dis} y_{a,t}^{es} \quad (33)$$

$$e_{a,\tau,t,s}^{es} = e_{a,\tau-1,t,s}^{es} + \eta^{ch} P_{a,\tau,t,s}^{es+} - P_{a,\tau,t,s}^{es-} / \eta^{dis} \quad (34)$$

$$e_{a,\tau,t,s}^{es} \Big|_{\tau=|T|} = e_{a,\tau,t,s}^{es} \Big|_{\tau=0} \quad (35)$$

$$P_{a,\tau,t,s}^{es+} \cdot P_{a,\tau,t,s}^{es-} = 0 \quad (36)$$

where $\bar{\lambda}_{a,\tau,t,s}^{pv}$ is the expected power generation of the PV with 1kW capacity at node a time τ in scenario s . $\gamma^{ch}, \gamma^{dis}$ are the charging and discharging rates of the ESS with 1kWh capacity. η^{ch}, η^{dis} are the charging and discharging efficiency of the ESS.

3) TN operation constraints: Firstly, the scheduled charging power at RCS a should be less than its capacity:

$$\sum_{i \in \mathcal{N}_m} P_{i,a,\tau,t}^{ch} \leq y_{a,t}^{cs}, \forall a \in A, \tau \in \mathcal{T}_t. \quad (37)$$

Besides, the shifted charging demand for DR at node i and time τ in the TN is defined as follows:

$$P_{i,\tau,t}^{sh} = \bar{d}_{i,\tau,t}^{ch} - \sum_{a \in A} P_{i,a,\tau,t}^{ch}, \forall i \in \mathcal{N}_m, \tau \in \mathcal{T}_t. \quad (38)$$

$P_{i,\tau,t}^{sh} > 0$ means the charging demand is unfulfilled while $P_{i,\tau,t}^{sh} < 0$ denotes the unfulfilled demand being recharged at time τ . $P_{i,\tau,t}^{sh}$ in the daily operation should satisfy:

$$\sum_{i \in \mathcal{N}_m} \sum_{\tau \in \mathcal{T}_t} P_{i,\tau,t}^{sh} \leq 0; P_{i,\tau,t}^{sh} \leq P_{i,\tau,t}^{sh} \leq \bar{P}_{i,\tau,t}^{sh} \quad (39)$$

to meet the total charging demand in one day and constrain $P_{i,\tau,t}^{sh}$ in a reasonable range due to limited charging flexibility.

D. Capacity-Dependent Uncertainty Set of PV Generation

The uncertainties in the PDN are mainly incurred by the loads and PVs. On the other hand, load forecasting is generally more accurate than renewable forecasting. Besides, it is decision independent and can be easily incorporated in the nodal power injections. Therefore, we focus on the uncertainty modeling of PVs.

The uncertainty source of PV generation is exogenous, which is mainly driven by weather conditions. However, in the planning model, the magnitude of PV power available at RCS depends on the installed PV capacity $y_{a,t}^{pv}$. Therefore, the capacity-dependent uncertainty of PV generation does not refer to the meteorological source itself, but to the capacity-scaled magnitude of PV uncertainty at RCS. A larger installed PV capacity implies a proportionally larger uncertain PV output range. By defining the random variable ξ^{pv} as PV generation uncertainty in magnitude, we have:

$$\tilde{P}_{a,\tau,t,s}^{pv} = \bar{\lambda}_{a,\tau,t,s}^{pv} y_{a,t}^{pv} - \xi_{a,\tau,t,s}^{pv}, \forall a \in A, \tau \in \mathcal{T}_t, s \in S. \quad (40)$$

To quantify ξ^{pv} in the operation period, the following BUS is constructed for the PV uncertainty in scenario s and stage t :

$$\Xi_{t,s}^{pv}(\mathbf{y}_t^{pv}) := \left\{ \begin{array}{l} \xi_{t,s}^{pv} \in \mathbb{R}^{|A| \times |\mathcal{T}_t|} : \\ -\alpha_{a,\tau,t,s}^{pv} \bar{\lambda}_{a,\tau,t,s}^{pv} y_{a,t}^{pv} \leq \xi_{a,\tau,t,s}^{pv} \leq \alpha_{a,\tau,t,s}^{pv} \bar{\lambda}_{a,\tau,t,s}^{pv} y_{a,t}^{pv} \\ \sum_{\tau \in \mathcal{T}_t} |\xi_{a,\tau,t,s}^{pv}| \leq \Gamma_{a,t,s}^{pv} \\ \Gamma_{a,t,s}^{pv} = \beta_{a,s}^{pv} \sum_{\tau \in \mathcal{T}_t} \bar{\lambda}_{a,\tau,t,s}^{pv} y_{a,t}^{pv} \end{array} \right. \quad (41)$$

where $\alpha_{a,\tau,t,s}^{pv}$ is the deviation factor. Parameter $\Gamma_{a,t,s}^{pv}$ models the temporal correlation of $\xi_{a,\tau,t,s}^{pv}$ and the spatial correlation is considered to be insignificant, which is reasonable in practice.

E. Planning Model under Decision Dependent Uncertainties

Combing the established DDU sets of Eq.(3) and (41), the original planning model under DDUs can be rewritten as:

$$V = \min \sum_t (1 + \gamma_{IR})^{1-t} \left(C_t^{inv} + \max_{\substack{\xi_{i,s}^{pv} \in \Xi_{i,s}^{pv} \\ \xi_{i,s}^{ch} \in \Xi_{i,s}^{ch}}} \left(C_t^{ope,pn} + C_t^{ope,ce} + C_t^{ope,m} \right) \right) \quad (42)$$

s.t. 1) Planning constraints:(9)–(20),

2) PDN operation constraints: (21)–(36),

3) TN operation constraints: (37)–(39).

To improve the computational tractability of (42), the following reformulation is established with a decomposable structure for the objective in (42):

$$V = \min \sum_t (1 + \gamma_{IR})^{1-t} \left(C_t^{inv} + V_t^{ope,pn} + V_t^{ope,ce} + V_t^{ope,m} \right) \quad (43)$$

$$V_t^{ope,pn} \geq \sum_{s \in S} D_s \left(\sum_{\tau \in \mathcal{T}_t} \left(C_{\tau,t,s}^{es} \sum_{a \in A} P_{a,\tau,t,s}^{es+} + C_{\tau,t,s}^{r} \sum_{a \in A} \left(\bar{\lambda}_{a,\tau,t,s}^{pv} y_{a,t}^{pv} - P_{a,\tau,t,s}^{pv} \right) \right) \right) + v_{t,s}^{mg} \quad (44)$$

$$v_{t,s}^{mg} = \max_{\xi_{i,s}^{pv} \in \Xi_{i,s}^{pv}} \sum_{\tau \in \mathcal{T}_t} C_{\tau,t,s}^{mg} \left(P_{\tau,t,s}^{mg} + \sum_{a \in A} \xi_{a,\tau,t,s}^{pv} \right) \quad (45)$$

$$V_t^{ope,ce} \geq \sum_{s \in S} D_s \max_{k=1,\dots,K} \left(c_{k,t}^{ce} \left(\max_{\xi_{i,s}^{pv} \in \Xi_{i,s}^{pv}} \sum_{\tau \in \mathcal{T}_t} C_{\tau,t,s}^{mg} \left(P_{\tau,t,s}^{mg} + \sum_{a \in A} \xi_{a,\tau,t,s}^{pv} \right) \right) \right) + b_{k,t}^{ce} \quad (46)$$

$$V_t^{ope,m} \geq \sum_{s \in S} D_s \sum_{\tau \in \mathcal{T}_t} (v_{\tau,t}^{dist} + v_{\tau,t}^{inc}) \quad (47)$$

$$v_{\tau,t}^{dist} \geq \sum_{i \in N_m} \max_{k=1,\dots,K} \left(\lambda_k^{dist} \sum_{a \in A} dist_{ia} P_{i,a,\tau,t}^{ch} + b_k^{dist} \right) \quad (48)$$

$$v_{\tau,t}^{inc} = \max_{\xi_{i,\tau,t}^{ch} \in \Xi_{i,\tau,t}^{ch}} \sum_{i \in N_m} \lambda^{inc} \left(P_{i,\tau,t}^{sh} + \xi_{i,\tau,t}^{ch} \right)^+ \quad (49)$$

The system-wide under-commitment PV generation (i.e., $\sum_{a \in A} \xi_{a,\tau,t,s}^{pv}$) is considered to be compensated by the main grid which is modeled by (45) with its corresponding carbon cost added in (46). Meanwhile, the deviation of charging demand (i.e., $\xi_{i,\tau,t}^{ch}, \forall i, \tau, t$) will influence the practical service quality of RCSs and is described by (49).

III. SOLUTION METHODOLOGY

A. Reformulation for PDN Model

To obtain a solvable form of the robust model with DDU and improve solving efficiency, the reformulations of (45), (46) and (49) are required.

Proposition 1: (45) is equivalent to the following equation:

$$v_{t,s}^{mg} = \sum_{\tau \in \mathcal{T}_t} \left(C_{\tau,t,s}^{mg} P_{\tau,t,s}^{mg} + \sum_{a \in A} \min(\alpha_{a,\tau,t,s}^{pv}, \beta_{a,\tau,t,s}^{pv}) \cdot C_{\tau,t,s}^{mg} \bar{\lambda}_{a,\tau,t,s}^{pv} y_{a,t}^{pv} \right). \quad (50)$$

Proof: Since the power deviations of PVs at different nodes are considered to be independent, the BUS of (41) can be modeled as $\Xi_{i,t,s}^{pv}(\mathbf{y}_t^{pv}) = \times_{a=1}^{|\mathcal{S}|} \Xi_{a,t,s}^{pv}(\mathbf{y}_{a,t}^{pv})$ (i.e., Cartesian product

of different nodes). Let $\xi_{a,\tau,t,s}^{pv} = \xi_{a,\tau,t,s}^{pv,+} - \xi_{a,\tau,t,s}^{pv,-}$ ($\xi_{a,\tau,t,s}^{pv,+/-} \geq 0$), (45) can be rewritten as

$$v_{t,s}^{mg} = \sum_{\tau \in \mathcal{T}_t} C_{\tau,t,s}^{mg} P_{\tau,t,s}^{mg} + \sum_{a \in A} \max_{\substack{\xi_{a,t,s}^{pv} \in \Xi_{a,t,s}^{pv} \\ \tau \in \mathcal{T}_t}} \sum_{\tau \in \mathcal{T}_t} C_{\tau,t,s}^{mg} \xi_{a,\tau,t,s}^{pv} \quad (51)$$

In the following, we omit the subscript a, t, s for simplicity and the dual problem of v^{dev} is equal to:

$$\min_{\pi_1^1, \pi_2^2, \pi_3^3} \sum_{\tau \in \mathcal{T}_t} \left(\pi_1^1 \alpha_{\tau}^{pv} + \pi_2^2 \alpha_{\tau}^{pv} + \pi_3^3 \beta_{\tau}^{pv} \right) \bar{\lambda}_{\tau}^{pv} y_{\tau}^{pv} \quad (52)$$

$$s.t. \pi_1^1 + \pi_3^3 \geq C_{\tau}^{mg} + \pi_2^2; \pi_2^2 + \pi_3^3 + C_{\tau}^{mg} \geq \pi_1^1$$

where $\pi_1^1, \pi_2^2, \pi_3^3$ are positive dual variables for constraints in BUS of (41). To minimize the right-hand side of (52), the dual variables with $\min(\alpha_{\tau}^{pv}, \beta_{\tau}^{pv})$ are set to $C_{\tau,t,s}^{mg}$ and others to 0, which completes the proof of Proposition 1.

Next, to deal with the blocked carbon emission cost under PV generation uncertainty in (46), we give the following proposition to obtain a tractable reformulation for (46).

Proposition 2. Given the BUS for PV uncertainty in (41), $V_t^{ope,ce}$ in (46) is equivalent to the linear program:

$$V_t^{ope,ce} \geq \sum_{s \in S} D_s \max_{k=1,\dots,K} \left(c_{k,t}^{ce} \left(\sum_{\tau \in \mathcal{T}_t} C_{\tau,t,s}^{mg} P_{\tau,t,s}^{mg} + \sum_{a \in A} \sum_{\tau \in \mathcal{T}_t} C_{\tau,t,s}^{mg} \cdot \min(\alpha_{a,\tau,t,s}^{pv}, \beta_{a,\tau,t,s}^{pv}) \bar{\lambda}_{a,\tau,t,s}^{pv} y_{a,t}^{pv} \right) \right) + b_{k,t}^{ce}. \quad (53)$$

Proof. Define the system-wide carbon emission as:

$$CE_{t,s} = \sum_{\tau \in \mathcal{T}_t} C_{\tau,t,s}^{mg} P_{\tau,t,s}^{mg} + \max_{\xi_{i,t,s}^{pv} \in \Xi_{i,t,s}^{pv}} C_{\tau,t,s}^{mg} \sum_{\tau \in \mathcal{T}_t} \sum_{a \in A} \xi_{a,\tau,t,s}^{pv}. \quad (54)$$

Similar to the proof of Proposition 1, the dual problem of the second term in (54) is:

$$\sum_{a \in A} \min_{\omega_{a,\tau}^1, \omega_{a,\tau}^2, \omega_{a,\tau}^3} \sum_{\tau \in \mathcal{T}_t} \left(\omega_{a,\tau}^1 \alpha_{a,\tau}^{pv} + \omega_{a,\tau}^2 \alpha_{a,\tau}^{pv} + \omega_{a,\tau}^3 \beta_{a,\tau}^{pv} \right) \bar{\lambda}_{a,\tau}^{pv} y_{a,t}^{pv} \quad (55)$$

$$s.t. \omega_{a,\tau}^1 + \omega_{a,\tau}^3 \geq C_{\tau}^{mg} + \omega_{a,\tau}^2; C_{\tau}^{mg} + \omega_{a,\tau}^2 + \omega_{a,\tau}^3 \geq \omega_{a,\tau}^1$$

Here, we omit the subscript t, s for simplicity and $\omega_{a,\tau}^1, \omega_{a,\tau}^2, \omega_{a,\tau}^3$ are positive dual variables for constraints in the BUS of (41). To minimize (55), the dual variables with $\min(\alpha_{a,\tau}^{pv}, \beta_{a,\tau}^{pv})$ are set to $C_{\tau,t,s}^{mg}$ and others to 0. Then, the system-wide carbon emission (54) is equivalent to:

$$CE_{t,s} = \sum_{\tau \in \mathcal{T}_t} C_{\tau,t,s}^{mg} P_{\tau,t,s}^{mg} + \sum_{a \in A} \sum_{\tau \in \mathcal{T}_t} C_{\tau,t,s}^{mg} \cdot \min(\alpha_{a,\tau,t,s}^{pv}, \beta_{a,\tau,t,s}^{pv}) \bar{\lambda}_{a,\tau,t,s}^{pv} y_{a,t}^{pv}$$

and we can further conclude Proposition 2.

B. Reformulation for TN Model

1) Linear Program Approximation Model

The optimization of (49) is difficult to solve due to the sum of inner maximization problem (i.e., adversarial problem). Meanwhile, the BUS of $\xi_{i,\tau,t}^{ch}$ is decision dependent. To obtain the tractable form of (49), the following lemma is established.

Lemma 1: The following optimization model

$$\begin{aligned}
 & \min_{\psi_{i,k} \geq 0, \nu \geq 0, \lambda_i^{+/-}, \gamma_i} \lambda^{inc} \left(\nu \beta^{ch} + \sum_{i \in \mathcal{N}_m} \gamma_i \right) \\
 & s.t., \lambda_i^+ + \psi_{i,k} \geq c_{i,k}, \forall i \in \mathcal{N}_m, k \in \{1, 2\} \\
 & \lambda_i^- + \psi_{i,k} + c_{i,k} \geq 0, \forall i \in \mathcal{N}_m, k \in \{1, 2\} \\
 & \gamma_i \geq \psi_{i,k} + d_{i,k}, \forall i \in \mathcal{N}_m, k \in \{1, 2\} \\
 & \theta_i^{pu} \nu \geq \lambda_i^+, \theta_i^{pu} \nu \geq \lambda_i^-, \forall i \in \mathcal{N}_m
 \end{aligned} \quad (56)$$

provides a conservative upper approximation of (49), where $c_{i,1} = \alpha_{i,\tau}^{ch} \bar{d}_{i,\tau,t}^{ch}$, $d_{i,1} = P_{i,\tau,t}^{sh}$, $c_{i,2} = 0$, $d_{i,2} = 0$. $\gamma_i, \nu, \lambda_i^{+/-}, \psi_{i,k}$ are the dual variables defined in (60).

Proof: Firstly, we reformulate the BUS (3) by defining a new random variable $\xi_{i,\tau,t}^{pu}$ with its BUS:

$$\Xi_t^{pu} := \left\{ \begin{array}{l} \xi_t^{pu} \in R^{|\mathcal{N}_m| \times |\mathcal{T}|} : \\ -1 \leq \xi_{i,\tau,t}^{pu} \leq 1 \\ \sum_{i \in \mathcal{N}_m} \theta_{i,\tau,t}^{pu} |\xi_{i,\tau,t}^{pu}| \leq \beta_\tau^{ch} \end{array} \right\} \quad (57)$$

where $\theta_{i,\tau,t}^{pu}$ is equal to $\alpha_{i,\tau}^{ch} \bar{d}_{i,\tau,t}^{ch} / \sum_{i \in \mathcal{N}_m} \bar{d}_{i,\tau,t}^{ch,0}$, which is an affine function of $\mathbf{y}_t^{cs}$. Then, $\xi_{i,\tau,t}^{ch} = \alpha_{i,\tau}^{ch} \bar{d}_{i,\tau,t}^{ch} \cdot \xi_{i,\tau,t}^{pu}$ and the right-hand side of (49) is equal to

$$h_{\tau,t}^{inc} = \lambda^{inc} \max_{\xi_t^{pu} \in \Xi_t^{pu}} \sum_{i \in \mathcal{N}_m} \max \left(P_{i,\tau,t}^{sh} + \alpha_{i,\tau}^{ch} \bar{d}_{i,\tau,t}^{ch} \cdot \xi_{i,\tau,t}^{pu}, 0 \right). \quad (58)$$

Hence, (58) is converted into the mixed-integer program:

$$\begin{aligned}
 h_{\tau,t}^{inc} &= \max \lambda^{inc} \sum_{i \in \mathcal{N}_m} \sum_{k=1}^2 z_{i,k} (c_{i,k} \xi_{i,\tau,t}^{pu} + d_{i,k}) \\
 s.t., z_{i,k} &\in \{0, 1\}, \sum_{k=1}^2 z_{i,k} = 1, \xi_t^{pu} \in \Xi_t^{pu} \\
 c_{i,1} &= \alpha_{i,\tau}^{ch} \bar{d}_{i,\tau,t}^{ch}, d_{i,1} = P_{i,\tau,t}^{sh}, c_{i,2} = 0, d_{i,2} = 0
 \end{aligned} \quad (59)$$

To circumvent the difficulty of maximizing the nonlinear program of (59), we utilize the linearization method and duality theorem. In the following, we omit the subscript τ, t, s for simplicity since these indices have no effect on the derivation. Besides, replacing the random variable ξ_i^{pu} by $\xi_i^{pu} = \xi_i^+ - \xi_i^-$, $\xi_i^+, \xi_i^- \geq 0$ and letting $\Delta_{i,k}^+ = z_{i,k} \xi_i^+$, $\Delta_{i,k}^- = z_{i,k} \xi_i^-$, (59) can be rewritten as:

$$\begin{aligned}
 & \max_{\xi_i^+, \xi_i^- \geq 0, z_{i,k} \in \{0, 1\}} \lambda^{inc} \left(\sum_{i \in \mathcal{N}_m} \sum_{k=1}^2 c_{i,k} (\Delta_{i,k}^+ - \Delta_{i,k}^-) + d_{i,k} z_{i,k} \right) \\
 & s.t., \sum_{k=1}^2 z_{i,k} = 1, \forall i: \gamma_i \\
 & \sum_{i \in \mathcal{N}_m} \theta_i^{pu} (\xi_i^+ + \xi_i^-) \leq \beta^{ch} : \nu \\
 & \Delta_{i,k}^+ = z_{i,k} \xi_i^+, \Delta_{i,k}^- = z_{i,k} \xi_i^-, \forall i: \lambda_i^{+/-} \\
 & \Delta_{i,k}^+ + \Delta_{i,k}^- + s_{i,k} = z_{i,k}, \forall i, k: \psi_{i,k}
 \end{aligned} \quad (60)$$

where $s_{i,k}$ is a positive slack variable. By *fractional relaxation* of binary variable $z_{i,k}$, we can derive the dual minimization problem of (60), which is equal to (56). Since (56) is derived by *fractional relaxation*, it gives a conservative approximation of

(49), which concludes **Lemma 1**.

Remark 1. Although Lemma 1 can only give an upper bound for (60), it provides a linear program reformulation and has a better convergence rate. Moreover, it facilitates designing the customized algorithm to deal with the bilinear term caused by decision dependent parameters (i.e., $\theta_{i,\tau,t}^{pu}, c_{i,k}, d_{i,k}$) in (56).

Proposition 3. Let $\hat{\mathbf{x}}, \hat{V}(\hat{\mathbf{x}})$ be the optimal solution and objective value obtained from the relaxed problem of (42), and $\mathbf{x}^*, V(\mathbf{x}^*)$ be the optimal solution and objective value for the original problem (42). Then, we have:

$$V(\mathbf{x}^*) \leq V(\hat{\mathbf{x}}) \leq \hat{V}(\hat{\mathbf{x}}) \leq \hat{V}(\mathbf{x}^*) \leq \gamma^{\text{relax}} V(\mathbf{x}^*). \quad (61)$$

where γ^{relax} is the integrality gap coefficient.

Proposition 3 characterizes the fractional relaxation gap through γ^{relax} . However, deriving a closed-form bound for γ^{relax} is generally challenging for the proposed model, since the adversarial problem of (60) is coupled with both planning variables and TN operation variables. This difficulty is consistent with linear program relaxation-based approximation methods, where exactness can usually be established only under special structural conditions [34]. Therefore, instead of claiming a general theoretical bound, we quantify the practical magnitude of this gap through experimental evaluation in the case studies in Section IV.B.2.

2) Customized Algorithm for Solving Bilinear Term

In (56), there exists bilinear term $\theta_i^{pu} \nu$, which remains hard to solve. To tackle this difficulty, we propose a customized algorithm combined with ADM [35] and CCP [36]. Specifically, let Θ_{prim} is the set of all decision variables except for ν , and $\Phi_{\text{dual}} = \{\nu\}$. Then, the optimization problem is divided into two iteration loops, namely, *inner ADM loop* and *outer CCP loop*.

The inner ADM loop contains two steps at s -th iteration:

1) Solve the following optimization problem with the given solution $\Theta_{\text{prim}}^{s-1}$ at $(s-1)$ -th iteration to obtain Φ_{dual}^s :

$$\begin{aligned}
 & \min_{\nu \geq 0} \nu \beta^{ch} \\
 & s.t., (\theta_i^{pu})^{s-1} \nu \geq (\lambda_i^+)^{s-1}, (\theta_i^{pu})^{s-1} \nu \geq (\lambda_i^-)^{s-1}, \forall i \in \mathcal{N}_{tn}
 \end{aligned} \quad (62)$$

2) Solve the relaxed problem (42) with the reformulation (56) and the given solution Φ_{dual}^s to obtain Θ_{prim}^s and $\hat{V}(\hat{\mathbf{x}}^s)$.

3) Check the inner loop convergence criterion.

When inner loop convergence criterion is satisfied for k -th outer loop, we obtain a local minimum and denote this optimal inner loop solution as $\tilde{\mathbf{x}}^k$. To improve the solution from the inner loop, we firstly rewrite the non-convex constraints of (56) in the form of DC function as follows:

$$(\lambda_i^{+/-} + 1)^2 + (\theta_i^{pu} - \nu)^2 \leq (\theta_i^{pu} + \nu)^2 + (\lambda_i^{+/-} - 1)^2. \quad (63)$$

In addition, linearize the right-hand side of (63) at $\tilde{\mathbf{x}}^k$ by:

$$\begin{aligned} f_1^{ccp}(\theta_i^{pu}, \nu) &\triangleq \left((\tilde{\theta}_i^{pu})^k + (\tilde{\nu})^k \right)^2 + 2 \left((\tilde{\theta}_i^{pu})^k + (\tilde{\nu})^k \right) \left(\theta_i^{pu} + \nu - (\tilde{\theta}_i^{pu})^k - (\tilde{\nu})^k \right) \\ f_2^{ccp}(\lambda_i^{+/-}) &\triangleq \left((\tilde{\lambda}_i^{+/-})^k - 1 \right)^2 + 2 \left((\tilde{\lambda}_i^{+/-})^k - 1 \right) \left(\lambda_i^{+/-} - (\tilde{\lambda}_i^{+/-})^k \right), \forall i \in \mathcal{N}_m \end{aligned} \quad (64)$$

Then, the DC constraints in (56) are convexified as:

$$\left(\lambda_i^{+/-} + 1 \right)^2 + \left(\theta_i^{pu} - \nu \right)^2 \leq f_1^{ccp}(\theta_i^{pu}, \nu) + f_2^{ccp}(\lambda_i^{+/-}), \quad (65)$$

and the relaxed problem of (42) can be solved with QC (65) to obtain the solution $\tilde{\mathbf{x}}^k$. If $\|\tilde{\mathbf{x}}^k - \mathbf{x}^k\| < \varepsilon^{out}$, it indicates $\tilde{\mathbf{x}}^k$ is the optimal value for the relaxed problem of (42). Otherwise, let $\hat{\mathbf{x}}_{init}^{k+1} = \tilde{\mathbf{x}}^k$ be the initial solution for $(k+1)$ -th iteration and go to the next iteration. The combined ADM-CCP algorithm is summarized as follows:

Algorithm 1: Combined ADM-CCP algorithm

Input: Initial $k=0$, $s=0$ and $\hat{\mathbf{x}}_{init}^0$. Set inner and outer loop tolerance: $\varepsilon^{in}, \varepsilon^{out}$. The maximum iteration is K^{max} .

for $k=0, 1, \dots, K^{max}$ **do**:

- (1) $s=0$, $\hat{\mathbf{x}}^{0,k} = \{\Theta_{prim}^{0,k}, \Phi_{dual}^{0,k}\} = \hat{\mathbf{x}}_{init}^k$. calculate $\hat{V}(\hat{\mathbf{x}}^{0,k})$.
- (2) **While** $\|\hat{V}(\hat{\mathbf{x}}^{s,k}) - \hat{V}(\hat{\mathbf{x}}^{s-1,k})\| \geq \varepsilon^{in}$ or $s=0$ **do**:
 - 1: $s=s+1$.
 - 2: Solve (62) with $\Theta_{prim}^{s-1,k}$ to obtain $\Phi_{dual}^{s,k}$.
 - 3: Solve the relaxed problem of (42) with the given solution $\Phi_{dual}^{s,k}$ to obtain $\Theta_{prim}^{s,k}$.
 - 4: Record $\hat{\mathbf{x}}^{s,k} = \{\Theta_{prim}^{s,k}, \Phi_{dual}^{s,k}\}$; Calculate $\hat{V}(\hat{\mathbf{x}}^{s,k})$.
- (3) Let $\tilde{\mathbf{x}}^k = \hat{\mathbf{x}}^{s,k}$ and solve the relaxed problem of (42) with QC (65) to obtain the solution $\tilde{\mathbf{x}}^k$.
- (4) **if** $\|\tilde{\mathbf{x}}^k - \mathbf{x}^k\| < \varepsilon^{out}$:
- (5) $\hat{\mathbf{x}} = \tilde{\mathbf{x}}^k$; **break**.
- (6) **else**:
- (7) $\hat{\mathbf{x}}_{init}^{k+1} = \tilde{\mathbf{x}}^k$; go to Step (1).

Output: the optimal solution $\hat{\mathbf{x}}$.

Proposition 4. The sequence of objective function values, $\{\hat{V}(\Theta_{prim}^{s-1}, \Phi_{dual}^s), \hat{V}(\Theta_{prim}^s, \Phi_{dual}^s); s=1, 2, \dots\}$, generated by the inner loop of **Algorithm 1**, is *monotonically nonincreasing* and converges to a finite value, which is an upper bound for $\hat{V}(\hat{\mathbf{x}})$.

Proof. Note that $\{\Theta_{prim}^{s-1}, \Phi_{dual}^{s-1}\}$ is a feasible solution from $(s-1)$ -th iteration. For the inner subproblem in Step 2, one can always find a solution $\{\Theta_{prim}^{s-1}, \Phi_{dual}^s\}$ which satisfies $\hat{V}(\Theta_{prim}^{s-1}, \Phi_{dual}^s) \leq \hat{V}(\Theta_{prim}^{s-1}, \Phi_{dual}^{s-1})$. Next, in the inner subproblem in Step 3, since $\{\Theta_{prim}^{s-1}, \Phi_{dual}^s\}$ is a feasible solution for the relaxed problem of (42), we must have $\hat{V}(\Theta_{prim}^s, \Phi_{dual}^s)$ is no larger than $\hat{V}(\Theta_{prim}^{s-1}, \Phi_{dual}^s)$ by optimizing Θ_{prim} . Therefore, $\{\hat{V}(\Theta_{prim}^{s-1}, \Phi_{dual}^s), \hat{V}(\Theta_{prim}^s, \Phi_{dual}^s); s=1, 2, \dots\}$ is nonincreasing and the global optimal value $\hat{V}(\hat{\mathbf{x}})$ serves as a lower bound.

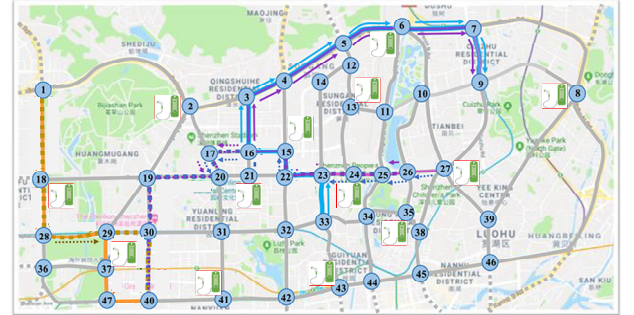
Proposition 5. The solution $\tilde{\mathbf{x}}^k$ generated by the outer loop of **Algorithm 1** makes the objective value $\hat{V}(\tilde{\mathbf{x}}^k) \leq \hat{V}(\tilde{\mathbf{x}}^k)$, served as a lower bound for $\hat{V}(\tilde{\mathbf{x}}^k)$.

Proof. Note that $\tilde{\mathbf{x}}^k$ is a feasible solution to the problem (42) from the last time of inner loop given $\tilde{\mathbf{x}}^k$. In the outer loop optimization, if $\|\tilde{\mathbf{x}}^{k-1} - \tilde{\mathbf{x}}^k\| \geq \varepsilon^{out}$, then $\hat{V}(\tilde{\mathbf{x}}^k) < \hat{V}(\tilde{\mathbf{x}}^k)$. Otherwise, the iteration terminates and $\hat{V}(\tilde{\mathbf{x}}^k) = \hat{V}(\tilde{\mathbf{x}}^k)$.

Remark 2. Compared to the conventional CCP algorithm, **Algorithm 1** accelerates the convergence as well as improving the solution accuracy by embedding the inner ADM loop.

IV. CASE STUDIES

In this section, the proposed model and algorithm are evaluated through several case studies. The simulation is performed in Python environment and solved by Gurobi with Intel Core i7-12700H CPU@ 2.30GHz and 16GB RAM.



TN-PDN pairs:
(2-39); (6-46); (8-27); (13-19); (15-15); (18-31); (21-10); (24-19); (27-23); (35-57); (37-35); (41-5); (43-65)

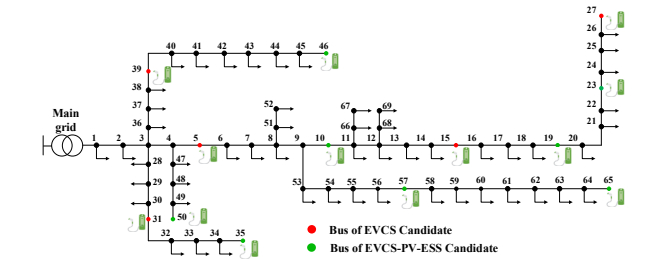


Fig. 4. The 47-node transportation system and 69-node distribution system.

A. System Description

The real-world TN system with 47 road intersections in Luohu district, Shenzhen city, China is used. The TN is coupled by a modified 69-bus distribution system as shown in Fig. 4. Besides, thirteen candidate nodes are chosen for RCS planning from 8 TN clusters based on clustering partition. The detailed clustering results, distance and topology matrices of the tested CPTN system are attached in [37].

Shenzhen, as a leading global city in the EV sector, its charging demand increases significantly in recent years with a large quantity of public EV charging data. Thus, we select several typical charging profiles with its expected values as the charging demands for different TN nodes [37].

The parameter settings of the base case are given as follows. The construction cost for one RCS varies from \$110e3 to \$150e3 and capacity expansion costs of fast charger, PV and ESS units range from \$110~140/kW, \$190~220/kW, \$100~120

/kWh, respectively, based on the land price of nodes. γ_{IR} is set by 15% considering the relatively short economic life and rapid technology turnover of fast charging infrastructure. The curtailment cost of PVs is \$0.2/kW and operational cost of ESSs is \$0.02/kW. The energy and blocked carbon prices of the main grid are derived from the electricity market of China Southern Power Grid in [37].

The parameters of Eq.(2) is set by: $\beta^{dist} = 1.8, \tau^{dist} = 0.8$ and $\kappa = 350$ based on results in [30]. The traveling cost for EV charging in the TN is \$1e-3/(km·kW) with an incremental rate of \$1e-3/(km·kW) applied to per block interval of 2000 (km·kW). The incentive price for shifting EV charging demands is \$0.15/kW. The temporal charging flexibility in the benchmark case is $P_{i,\tau,t}^{sh} \in [-0.6\bar{d}_{i,\tau,t}^{ch}, 0.3\bar{d}_{i,\tau,t}^{ch}]$.

For the uncertainty settings, the deviation rate of PVs $\alpha_{b,\tau,s}^{pv}$ is distributed from 10% to 18% depending on different time periods in different scenarios. The budgeted parameter of PVs $\beta_{b,s}^{pv}$ is uniformly distributed from 8% to 15% for different nodes. The deviation rate of charging demand $\alpha_{i,\tau}^{ch}$ is distributed from 14% to 23% and β_{τ}^{ch} distributed from 10% to 17% for different time periods in one day.

The expected growth rate of charging demand is set at 25% considering the fast EV increase in Shenzhen. The growth rate of loads is 5% and four representative days of conventional loads are considered at each stage, which is similar to [19].

The time span of each period in the proposed model can be adjusted according to the practical needs of the system operator. In this paper, it is set to one year and the planning horizon is set as 5 years. Each typical scenario consists of 90 days.

B. Comparative Tests under Different Methods

B.1 Optimization results under different planning strategies

To demonstrate the significance of modeling the DDU of charging demand, the proposed model (Strategy I) is compared with the following models:

Strategy II: the robust model where the DDU set of (3) is modeled based on a newly defined binary variable $b_{i,t}^{cs} \in \{0,1\}$, which equals 1 if $y_{i,t}^{cs} \geq 500$ kW, and 0 otherwise. The detailed model is derived in Appendix. A.

Strategy III: the robust model without considering the impact of DDCDT, i.e., DIU model.

Strategy IV: the stochastic model where the DDU set of (3) is replaced by N_s scenarios for charging demand deviation (i.e., $\{\hat{\xi}_{i,s}^{ch}\}_{s=1}^{N_s}$) and incentive cost of (49) by the sample average:

$$v_{\tau,t}^{inc} = \frac{1}{N_s} \sum_{s=1}^{N_s} \sum_{i \in N_{in}} \lambda^{inc} (P_{i,\tau,t}^{sh} + \hat{\xi}_{i,\tau,t,s}^{ch})^+ \quad (66)$$

Strategy V: the proposed model without onsite ESSs.

The planning and operation costs are summarized in Table II.

TABLE II
PERFORMANCE COMPARISON UNDER DIFFERENT STRATEGIES

Strategy	I	II	III	IV	V
Total cost	61,152	61,276	61,307	60,355	61,794
Invest. cost	12,180	12,170	12,129	11,777	7,044
PDN cost	27,055	27,071	27,081	26,527	30,247

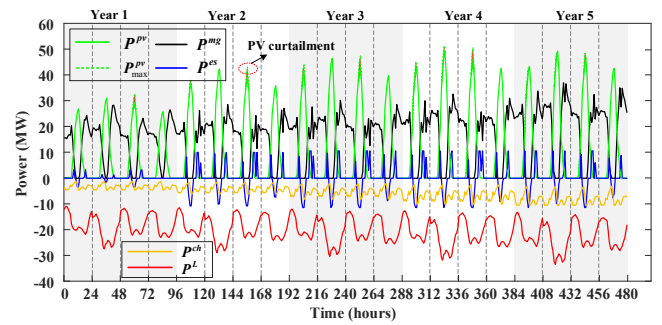
Carbon cost	15,711	15,730	15,738	15,478	18,258
TN cost	6,206	6,305	6,359	6,573	6,245
Capacity (MW)	10.6/47.	10.6/47.3	10.4/47.	10.1/46.8	10.4/35
	4/34.3	/34.4	3/34.1	/31.1	.8/0

Note: The unit for costs in the table is in thousands of US dollars. The results in the last row are the cumulative capacities of chargers, PVs and ESSs.

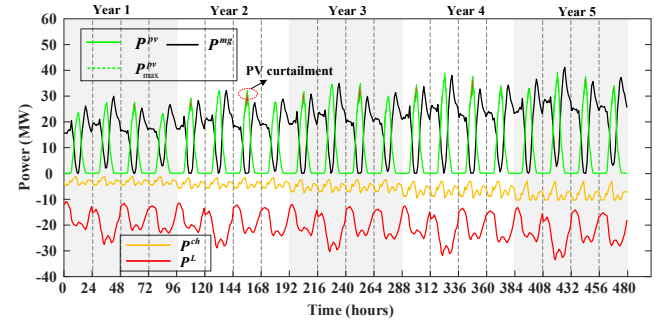
First, the comparison among Strategies I–III clearly reveals the value of modeling the “capacity-attracts-demand” feedback. Relative to Strategy III, Strategy II reduces the TN cost by 54 thousand dollars and the total cost by 31 thousand dollars, showing that incorporating decision dependence is beneficial. However, the proposed continuous DDU Strategy I further improves the results over Strategy II, reducing the TN cost by 99 thousand dollars and the total cost by 124 thousand dollars. This is because Strategy I gives a more accurate quantification of how staged capacity expansion continuously reshapes charging demand distribution. As a result, Strategy I achieves better spatial matching between charging demand and service supply and reduces both TN cost and overall system cost.

Second, although strategy IV has the lowest cost in expectation, its actual cost-efficiency varies under different charging demand profiles. Specifically, the actual TN cost of strategy IV fluctuates between 5.9~8.8 million dollars, which follows a heavy-tailed distribution. In contrast, strategy I is much more robust to the uncertainty of charging demand.

Thirdly, the contribution of on-site ESSs to carbon-aware EV integration can be assessed by comparing strategy I and V. Although the investment cost of strategy V decreased (due to sharp reduction of PVs and no ESSs), the PDN, carbon emission and TN costs rise significantly by 11.8%, 16.2% and 6.3%, respectively, comparing to those of strategy I. It indicates that the equipment of ESSs is crucial for the carbon-aware operation with high penetration of PVs. To further illustrate this point, the detailed scheduling results are shown in Fig. 5.



(a) The EVCS scheduling results with onsite ESSs



(b) The EVCS scheduling results without onsite ESSs

Fig. 5. The scheduling results with and without onsite ESSs.

P^{pv} , $P_{\max}^{pv}$, P^{mg} , P^{es} , P^{ch} , P^L in Fig. 5 are the vectors of PV output, maximum PV output, power from main grid, ESS charging power, charging demand and load demand. In Fig. 5(a), strategy I utilizes onsite ESSs to effectively store surplus energy of PVs during peak sunlight hours and discharges batteries at high energy price periods with no PV generation. In contrast, in Fig. 5(b), strategy V has to reduce PV installation but still encounters larger PV curtailment due to the absence of ESSs. Meanwhile, this also leads to a higher energy purchase cost and carbon emission cost from the main grid.

Additionally, Fig. 6 further illustrates the effect of DDCDT on charging demand redistribution. It can be observed that DDCDT does not uniformly alter the charging demand over all nodes and time periods. Instead, the redistribution is concentrated on a subset of nodes, i.e., the nodes with RCS construction and capacity expansion, directing SCP to the target

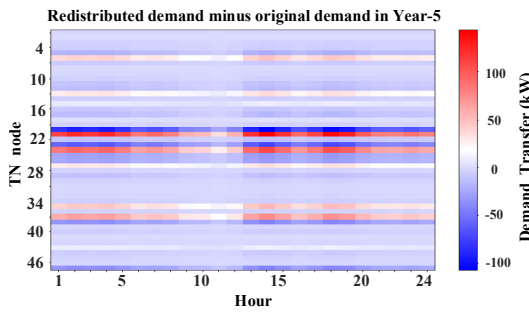


Fig. 6. Spatial-temporal charging demand transfer in Year 5.

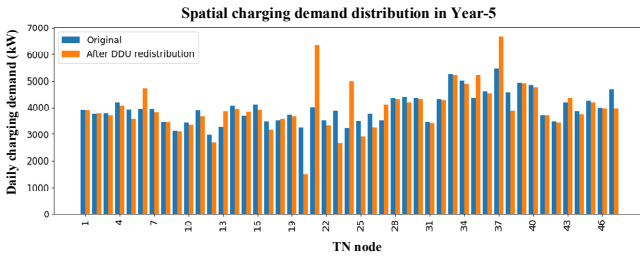


Fig. 7. Comparison of daily charging demand at each TN node before and after DDU redistribution in Year-5.

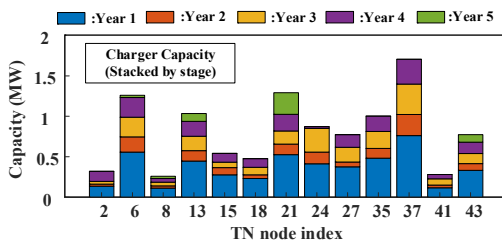


Fig. 8. Stage-wise charger planning results at different nodes.

deployment of RCS at nodes that exhibit both substantial charging demand and low grid stress.

Fig. 7 compares the daily charging demand at each TN node before and after charging demand redistribution. The results show that nodes with increased daily demand are accompanied by nodes with decreased demand, while the aggregate demand level is preserved. For example, nodes 21, 24, and 37 receive more charging demand after redistribution, whereas nodes 23, 25, and 38 experience demand reductions. This confirms that the proposed model improves the spatial matching between

charging demand and charging service by reallocating demand toward more attractive service locations.

The planning results of RCSs are further illustrated in Fig. 8, while the corresponding tabulated results are provided in [37]. It should be clarified that, in the tested system, all 13 candidate nodes are selected for initial RCS construction in Stage 1 since each candidate node has a positive Year-1 charger capacity. However, subsequent capacity expansion is selective and only a subset of the initially constructed stations receives additional charger capacity in later stages. The charger capacities of RCSs in Fig. 8 have a strongly positive correlation with local and nearby charging demands, which help to reduce the TN operation cost (e.g., TN node 37 has the largest capacity with highest charging demand). Meanwhile, we also find that a negative association between charger capacities and unit construction price exists (e.g., TN node 21 and 41 have comparable demands but the former has higher capacities due to the lower construction price of node 21).

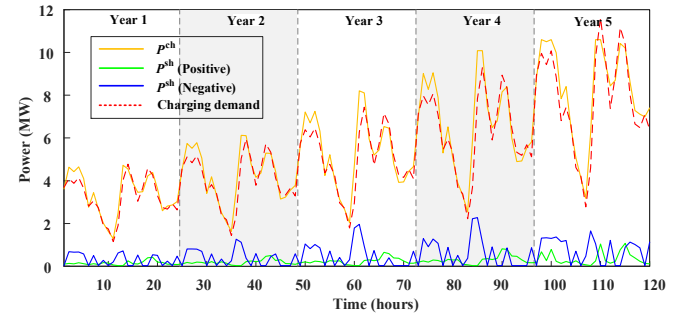


Fig. 9. The detailed EV scheduling results of the proposed model.

Observing the detailed EV scheduling results in Fig. 9, before year 5, the system operator shifts charging loads from evening peak load hours (6-8 p.m.) to lower energy price hours (0-6 a.m.) and peak sunlight hours to reduce energy, carbon and PV curtailment costs (working as a battery). However, in the last year operation, the scheduled charging power is smaller than charging demand in peak sunlight hours. This is because no additional PVs are constructed and EVs are controlled to reduce charging power (working as a generator).

B.2 Performance of proposed combined ADM-CCP algorithm

This subsection compares the performance of the proposed ADM-CCP algorithm to the single ADM [35] and single CCP method [36]. Meanwhile, we also add a comparison against one generic heuristic differential evolution (DE) algorithm [38]. For fairness, the DE algorithm only searches the same coupling variable $\Phi_{\text{dual}} = \{\nu\}$ as the procedure in (62), and each candidate is evaluated by solving the relaxed problem (42) with fixed $\Phi_{\text{dual}} = \{\nu\}$. The population of DE is 120 and the maximum generation is 40 to avoid unacceptable evaluating time since each individual in the DE algorithm entails the solution of planning model for fitness evaluation.

The performance results are summarized in Table III. The last column of Table III reports the empirical relaxation gap of the linear programming approximation of (56).

TABLE III
PERFORMANCE COMPARISON AMONG DIFFERENT METHODS

Method	TN incentive cost in (49)	Iteration time	Relaxation gap
Proposed method	\$2,781,980	1505 sec.	0.009%
ADM [35]	\$2,809,622	578 sec.	0.004%
CCP method [36]	\$2,782,015	2793 sec.	0.009%
DE algorithm [38]	\$3,025,456	35823 sec.	0.001%

First, although ADM converges fast, it terminates at a worse local optimum. In contrast, the proposed algorithm reduces 28 thousand dollars (1% TN incentive cost saving). Second, the proposed method significantly reduces computational time (46% in total) compared to CCP method by adding the inner ADM loop. Third, the heuristic DE method fails to achieve solution quality comparable to the proposed method within an acceptable computational budget. Specifically, the DE result is 8.8% higher in TN incentive cost than the proposed method, while requiring 23.8 times longer computation time. Therefore, the generic heuristic search becomes evaluation-dominated and computationally unattractive. In contrast, the proposed ADM-CCP algorithm exploits the mathematical structure of the reformulated problem, and only applies iterative approximation to the necessary bilinear block, thereby significantly improving solution quality and computational efficiency.

Finally, the relaxation gaps $|V(\hat{x}) - \hat{V}(\hat{x})|/V(\hat{x})$ under all methods are smaller than 0.01% TN incentive cost, which validates the good performance of approximation model of (56). Although a general theoretical upper bound on γ^{relax} is difficult to establish for the proposed continuous DDU model, the observed relaxation error is negligible in the case studies and is within an acceptable range for planning-level decision analysis.

C. Performance Analysis

C.1 Impact of DDCDT intensity

In this subsection, we test how the planning and scheduling strategy changes with different intensity of DDCDT controlled by hyperparameters $\tau^{dist}, \beta^{dist}, \kappa$. When testing one parameter, the other two are fixed at their baseline values. Table IV and Fig. 10 report the summarized results and the detailed sensitivity results are provided in [37].

Param.	Range	Total cost	Invest. cost	TN cost
τ^{dist}	0.2→1. 4	61,280→60,99 8	12,130→12,20 0	6,334→6,054
β^{dist}	1.5→3. 5	60,870→61,25 4	12,213→12,17 1	5,917→6,312
κ	200→5 00	61,185→61,11 1	12,138→12,17 6	6,235→6,170

Note: The unit for costs in the table is in thousands of US dollars.

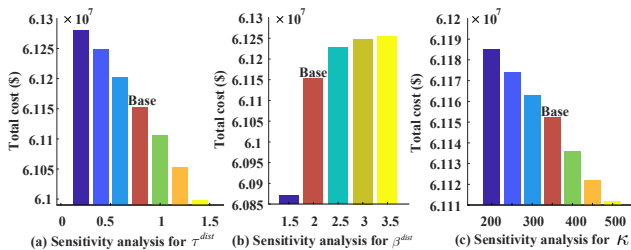


Fig. 10. Sensitivity analysis of hyperparameters of DDCDT model.

It should be noted that the impacts of $\tau^{dist}, \beta^{dist}, \kappa$ are mainly reflected in the investment cost and TN cost since these parameters enter the distance-based transfer matrix of Eq.(1) and (2).

As τ^{dist} increases from 0.2 to 1.4, the effect of DDCDT is enhanced, which improves the spatial matching between charging demand and service supply. Therefore, a significant reduction of TN cost (mainly from traveling cost reduction) is observed. Meanwhile, the cumulative charger capacity also increases from 10.4MW to 10.8MW to improve the real-time service level of RCSs, resulting in a higher investment cost. Second, increasing κ from 200 to 500 slightly reduces the total cost from 61.185 to 61.111 million USD, showing a weak and approximately linear sensitivity to the exponential decay scale.

In contrast, β^{dist} exhibits a nonlinear and saturating effect. Increasing β^{dist} from 1.5 to 3.5 imposes a stronger distance penalty and leads to higher total cost from 60.870 to 61.254 million USD and TN cost from 5.917 to 6.312 million USD. However, the marginal increase becomes much smaller as β^{dist} grows, indicating that the distance penalty gradually saturates after long-distance demand transfer has been sufficiently suppressed.

C.2 Impact of EV charging flexibility

Next, we change the flexible range of $P_{i,\tau,t}^{sh}$ from $\pm 10\% \bar{d}_{i,\tau,t}^{ch}$ to $\pm 60\% \bar{d}_{i,\tau,t}^{ch}$ to explore its impact on the planning and operational costs. The results are summarized in Table V.

EV flexible range	Total cost	Invest. cost	PDN cost	Carbon cost	TN cost
$[-0.2\bar{d}_{i,\tau,t}^{ch}, 0.1\bar{d}_{i,\tau,t}^{ch}]$	61,318	12,316	27,167	15,731	6,104
$[-0.4\bar{d}_{i,\tau,t}^{ch}, 0.2\bar{d}_{i,\tau,t}^{ch}]$	61,220	12,226	27,116	15,724	6,154
$[-0.6\bar{d}_{i,\tau,t}^{ch}, 0.3\bar{d}_{i,\tau,t}^{ch}]$	61,152	12,180	27,055	15,711	6,206
$[-0.8\bar{d}_{i,\tau,t}^{ch}, 0.4\bar{d}_{i,\tau,t}^{ch}]$	61,085	12,090	27,047	15,695	6,254
$[-\bar{d}_{i,\tau,t}^{ch}, 0.5\bar{d}_{i,\tau,t}^{ch}]$	61,025	12,183	26,970	15,583	6,289
$[-1.2\bar{d}_{i,\tau,t}^{ch}, 0.6\bar{d}_{i,\tau,t}^{ch}]$	60,971	12,294	26,868	15,481	6,328

Note: The unit for costs in the table is in thousands of US dollars.

First, the investment cost reaches a minimum at 40% flexible range and then increases. This is because the charger capacities rise linearly to avoid large traveling and incentive costs with charging flexibility increasing, while the capacity reduction of ESS tapers gradually for PV-peak absorption and uncertainty hedging. Besides, we can observe that the PDN operation and carbon emission costs decrease monotonically which validates the EV charging flexibility can facilitate load shifting into low energy price and CI periods as shown in Fig. 9.

Second, for the TN operation, the traveling cost is reduced since charging flexibility enlarges the temporal admissible charging windows of EVs so that the nearest RCS becomes feasible without spatial detours and drivers can “wait a little, charge nearby” instead of “drive farther, charge now”. On the other hand, the incentive cost gets increased to encourage more

DR. Consequently, the faster-rising incentive cost lifts the total TN operation cost with larger charging flexibility.

Finally, the total cost decreases monotonically with the increase of EV charging flexibility (by 0.35 million dollars from 10% to 60%) and shows a clear decreasing marginal benefit (The first 10% brings 98 thousand dollars in savings, while the last 10% only 53 thousand dollars).

Thereby, we recommend that the planner design suitable incentive DR programs or carbon-aware tariffs to encourage at least 30% EV users to participate into coordinated charging.

C.3 Impact of charging demand uncertainty

To explore the impact of charging demand uncertainty, we introduce an adjustable coefficient κ_{dev} to control the uncertain level of charging demand. Then, the varying deviation rates are given by $\tilde{\alpha}_{i,\tau}^{ch} = \kappa_{dev} \alpha_{i,\tau}^{ch}$ and $\tilde{\beta}_{\tau}^{ch} = \kappa_{dev} \beta_{\tau}^{ch}$. The results of sensitivity analysis of charging demand uncertainty are summarized in Table VI.

TABLE VI
COST RESULTS FOR THE IMPACT OF CHARGING DEMAND UNCERTAINTY

κ_{dev}	Total cost	Invest. cost	PDN cost	Carbon cost	TN cost
0.1	58,435	12,008	26,872	15,508	4,048
0.5	59,611	12,066	27,055	15,685	4,806
1	61,152	12,180	27,055	15,711	6,206
1.5	62,736	12,157	27,046	15,717	7,816
2	64,368	12,147	27,037	15,709	9,475
2.5	66,022	12,149	27,069	15,717	11,087

Note: The unit for costs in the table is in thousands of US dollars.

First, the total cost increases gradually from 58.4 million to 66 million dollars as κ_{dev} changes from 0.1 to 2.5. Moreover, the growth rate of total cost gets faster with the higher uncertain level of charging demand. Meanwhile, when $\kappa_{dev} \leq 1$, each part of cost increases to offset higher uncertainty of charging demand and avoid expensive DR cost from (49). In contrast, when $\kappa_{dev} > 1$, the cost increase is mainly caused by the rise of DR incentive from TN operation cost. In other words, the marginal price of capacity planning and PDN scheduling is larger than that of DR incentive after $\kappa_{dev} > 1$.

C.4 Impact of interest rate γ^{IR}

Recent studies of fast CS upgradation emphasize that fast charging technologies evolve rapidly and that discounting is necessary when evaluating intertemporal CS planning decisions. For example, authors in [39] adopt an annual discount rate of 10% for charging station upgrade analysis. To further examine the influence of this parameter, γ^{IR} is varied from 5% to 15% and the results are summarized in Table VII.

TABLE VII
COST RESULTS FOR THE IMPACT OF INTEREST RATE

γ^{IR}	Total cost	Invest. cost	PDN cost	Carbon cost	TN cost
0.050	71,438	13,362	31,833	18,558	7,686
0.075	68,512	13,062	30,460	17,729	7,261
0.100	65,881	12,404	29,433	17,176	6,868
0.125	63,414	12,285	28,194	16,414	6,521
0.150	61,152	12,180	27,055	15,711	6,206

Note: The unit for costs in the table is in thousands of US dollars.

As shown in Table VII, both the total cost and each cost component reduce as γ^{IR} increases. This trend should be interpreted as a larger discount rate reducing the present-value weight of future-stage costs, rather than as evidence that a higher interest rate is economically preferable. Despite the change in discounted cost level, the RCS planning structure remains stable. The number of selected RCSs in the final planning stage remains 13 for all tested interest rates, and the total RCS capacity varies only slightly from 10.538 MW to 10.545 MW. This indicates that the main planning conclusions are not driven by the specific choice of γ^{IR} .

D. Scalability discussion

D.1 Simulation result analysis

In this subsection, all 47 TN nodes are considered as candidate charging station locations, and each TN node is paired with one dedicated PDN node. The DDU model with 47 candidates is denoted as Strategy VI, while the DIU model with 47 candidates is denoted as Strategy VII. The results are summarized in Table VIII.

TABLE VIII
COMPUTATIONAL PERFORMANCE UNDER ENLARGED CANDIDATE SETS

Strategy	Total cost	Invest. cost	PDN cost	Carbon cost	TN cost	Comp time
I	61,152	12,180	27,055	15,711	6,206	0.42h
VI	60,108	12,572	26,992	15,665	4,879	7.72h
VII	60,233	12,520	27,019	15,692	5,002	0.89h

Note: The unit for costs in the table is in thousands of US dollars.

Compared with Strategy I, which restricts the planning model to 13 pre-selected TN nodes, Strategies VI and VII search over the full set of 47 TN nodes. The optimized layout contains 16 RCS sites distributed across the 8 TN clusters, with each cluster hosting 1–3 stations (detailed results are provided in [37]). The enlarged candidate set reduces the total cost mainly by lowering the TN cost, although it moderately increases the investment cost.

However, the full-candidate search significantly increases the computational burden. The computation time rises from 0.42 hour in Strategy I to 0.89 hour in Strategy VII and 7.72 hours in Strategy VI. This scalability issue is more pronounced for Strategy VI since the DDU formulation is solved iteratively within the CCP framework.

D.2 Potential directions for acceleration

A practical acceleration strategy is to first partition the TN into coherent clusters and then build a reduced candidate set by selecting low-cost and representative nodes from each cluster. This screens out redundant nearby candidates before optimization, thereby shrinking the search space while preserving spatial coverage.

Another possible strategy is motivated by the observation that Strategies VI and VII obtain the same RCS siting pattern, although their installed capacities differ. This indicates that the DDCDT mechanism mainly affects continuous sizing rather than discrete siting in this case. Thereby, an efficient acceleration scheme is that the DIU model is first solved to identify the optimal RCS locations. Then, the DDU model is solved with fixed siting variables to optimize station capacities.

This retains the benefit of full-network siting exploration while substantially reducing the computational burden of the iterative DDU solution.

V. CONCLUSION AND FUTURE WORK

This article proposes a carbon-aware robust planning model of RCSs with continuous DDU of charging demand. A customized combined ADM and CCP algorithm is developed to solve the established model efficiently with convergence guarantee. The key findings are summarized as follows:

1) The modeling of DDU improves the spatial matching between charging demand and service supply, which achieves 2.5% TN cost saving compared to the DIU model.

2) The proposed combined ADM-CCP algorithm makes a better tradeoff between solution quality and computational efficiency. Besides, the dual gap of proposed linear program approximation model is less than 0.01%, evidencing the near-lossless model fidelity.

3) The sensitive analysis shows that strengthening DDCDT intensity and enlarging EV charging flexibility systematically lower the total cost by redirecting load to nearby locations and attenuating concurrent charging peaks. In contrast, the rise of uncertainty level of charging demands leads to significant cost increase of TN operation.

In future work, we plan to derive theoretical bounds on the fractional relaxation gap by exploiting problem-specific monotonicity and tighter valid inequalities. Additionally, the semidefinite programming approximation with efficient computation algorithm can be explored to further tighten the relaxation gap.

APPENDIX

A. Detailed Model of Strategy II with binary decision variables DDU

To compare the strategy with binary decision variables DDU, we construct a DDU robust set related to a newly defined binary variable $b_{i,t}^{cs} \in \{0,1\}$, which equals 1 if $y_{i,t}^{cs} \geq 500$ kW, and 0 otherwise. The detailed model is described as follows.

First, the expected charging demand is modeled by a linear function of RCS siting variable, which is similar to (1):

$$\bar{d}_{\tau,t}^{ch}(\mathbf{b}_t^{cs}) = \bar{d}_{\tau,t}^{ch,0} + B_{\tau,t}^{tr} \mathbf{L}^tr \mathbf{b}_t^{cs}; \forall \tau \in \mathcal{T}, t \in T. \quad (67)$$

where $B_{\tau,t}^{tr}$ is the scaling factor of binary variables $b_{i,t}^{cs}$. In the case studies, $B_{\tau,t}^{tr} = 500\alpha_{\tau}^{tr}$. Furthermore, to differentiate DDU set of binary siting variable from the proposed one, we refer to [40] and construct the following DDU set to encompass all candidate probability distribution functions (PDFs) for the uncertain EV charging demand:

$$\mathcal{P}_{\tau,t}^{ch}(\mathbf{b}_t^{cs}) := \left\{ \begin{aligned} & \omega_{i,\tau}^{ch} \in R_+^{K_{i,\tau}^{ch}} \\ & \sum_{k \in K_{i,\tau}^{ch}} \omega_{i,\tau,k}^{ch} = 1 : (\bar{\lambda}_{i,0}, \underline{\lambda}_{i,0}) \\ & \bar{d}_{i,\tau,t}^{ch}(\mathbf{b}_t^{cs}) - \varepsilon_{i,\tau}^{\mu} \leq \sum_{k \in K_{i,\tau}^{ch}} \omega_{i,\tau,k}^{ch} d_{i,\tau,t,k}^{ch} \\ & \left[\bar{d}_{i,\tau,t}^{ch}(\mathbf{b}_t^{cs}) + \varepsilon_{i,\tau}^{\mu} : (\bar{\lambda}_{i,i}, \underline{\lambda}_{i,i}) \right] \end{aligned} \right\} \quad (68)$$

where $d_{i,\tau,t,k}^{ch}$ is the k -th possible realization of uncertain charging demand $\tilde{d}_{i,\tau,t}^{ch}$ at i -th TN node, τ -th operation time and t -th planning stage. $\omega_{i,\tau}^{ch} = \left\{ \omega_{i,\tau,k}^{ch} \right\}_{k=1}^{K_{i,\tau}^{ch}}$ represents the probabilities assigned to different realizations, whose total sum should be equal to 1. Parameters $\varepsilon_{i,\tau}^{\mu}$ determine the robustness of the DDU set. The variable in the bracket after each constraint corresponds to the dual variable of the constraint.

Based on the above DDU set defined by (68), the TN operation cost of Eq.(49) is modified by:

$$v_{\tau,t}^{inc} = \sup_{\mathbb{P}_{\tau,t}^{ch} \in \mathcal{P}_{\tau,t}^{ch}(\mathbf{b}_t^{cs})} \mathbb{E} \left[\sum_{i \in N_m} \lambda^{inc} \left(P_{i,\tau,t}^{sh} + (\tilde{d}_{i,\tau,t}^{ch} - \bar{d}_{i,\tau,t}^{ch})^+ \right) \right] \quad (69)$$

Furthermore, based on **Theorem 1** in [40], Eq.(69) can be reformulated as:

$$v_{\tau,t}^{inc} = \min_{\lambda, \underline{\lambda} \geq 0} \left(\begin{aligned} & \sum_{i \in N_m} (\bar{\lambda}_{i,0} - \underline{\lambda}_{i,0}) \\ & + \sum_{i \in N_m} (\bar{\lambda}_{i,i} (\bar{d}_{i,\tau,t}^{ch} + \varepsilon_{i,\tau}^{\mu}) - \underline{\lambda}_{i,i} (\bar{d}_{i,\tau,t}^{ch} - \varepsilon_{i,\tau}^{\mu})) \end{aligned} \right) \quad (70)$$

s.t. $(\bar{\lambda}_{i,0} - \underline{\lambda}_{i,0}) + (\bar{\lambda}_{i,1} - \underline{\lambda}_{i,1}) d_{i,\tau,t,k}^{ch} \geq \lambda^{inc} (P_{i,\tau,t}^{sh} + d_{i,\tau,t,k}^{ch} - \bar{d}_{i,\tau,t}^{ch})^+$

There exist nonlinear terms in objective functions due to the decision dependent $\bar{d}_{i,\tau,t}^{ch}(\mathbf{b}_t^{cs})$. However, for bilinear $\mathbf{b}_t^{cs}$, we can obtain the exact MILP reformulation of objective in (70) using McCormick envelopes. First, rewrite objective in (70) as:

$$v_{\tau,t}^{inc} = \min_{\lambda, \underline{\lambda} \geq 0} \left(\begin{aligned} & \sum_{i \in N_m} (\bar{\lambda}_{i,0} - \underline{\lambda}_{i,0}) \\ & + \sum_{i \in N_m} (\bar{\lambda}_{i,i} (\bar{d}_{i,\tau,t}^{ch,0} + \varepsilon_{i,\tau}^{\mu}) - \underline{\lambda}_{i,i} (\bar{d}_{i,\tau,t}^{ch,0} - \varepsilon_{i,\tau}^{\mu})) \\ & + B_{\tau,t}^{tr} \sum_{i \in N_m} (\bar{\lambda}_{i,i} (\mathbf{L}^{tr} \mathbf{b}_t^{cs})_{[i]} - \underline{\lambda}_{i,i} (\mathbf{L}^{tr} \mathbf{b}_t^{cs})_{[i]}) \end{aligned} \right) \quad (71)$$

where $(\mathbf{L}^{tr} \mathbf{z}_t^{cs})_{[i]}$ denotes i -th element of vector $\mathbf{L}^{tr} \mathbf{z}_t^{cs}$.

Introduce auxiliary variables: $u_{1,t,ij}^I = \bar{\lambda}_{i,i} z_{t,j}^{cs}$, $u_{2,t,ij}^II = \underline{\lambda}_{i,i} z_{t,j}^{cs}$, and insert them into (71) to obtain the linear reformulation. Additionally, the corresponding McCormick envelopes $M_{1,t,ij}^I, M_{1,t,ij}^{II}$ are added to define the feasible sets of auxiliary variables (details for this technique can be found in [41]).

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